CIE

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TITLE: SkillBridge
SCHOOL: Chitkara University, Rajpura, Punjab

DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO SHARE LOGIN AND PASSWORD WITH MENTORS

Name: Asmita Dey

Roll No/ Employee Code: 2210990193

All MEMBERS DETAILS INCLUDING SUPERVISOR

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ABSTRACT

1.1 Problem Background

In today's fast-paced digital environment, individuals often wish to contribute to social causes but face constraints such as limited time and lack of awareness about suitable opportunities. At the same time, organizations frequently require small, skill-based contributions that do not necessarily demand long-term commitment. Existing volunteering platforms largely focus on extended engagements and fail to support short-term, micro-level contributions effectively. This results in underutilization of skilled individuals and inefficiencies for organizations. Therefore, there is a clear need for a centralized, web-based platform that enables quick, skill-driven volunteering while ensuring proper tracking, validation, and recognition of contributions.

1.2 Objective of the Project

The primary objective of this project is to design and develop a web-based platform, *SkillBridge*, that facilitates short-duration, skill-based volunteering. The system aims to:

- Connect volunteers with relevant micro-tasks based on their skills.
- Enable organizations to post and manage tasks efficiently.
- Allow submission and verification of completed work.
- Provide recognition through points and badges.
- Maintain a structured and user-friendly environment for micro-volunteering.

1.3 Methods / Methodology

The proposed system follows a modular client-server architecture using modern web technologies. The platform consists of two primary user roles: volunteers and organizations. Users can register and log in securely, with role-based access controlling available features.

Organizations can create and manage tasks through a dedicated interface, specifying requirements, duration, and expected outcomes. Volunteers can browse available tasks and submit their work through text, links, or file uploads. A submission system enables organizations to review, approve, or reject completed tasks.

The system includes separate dashboards for volunteers and organizations. Volunteers can track their submissions, earned points, and badges, while organizations can monitor tasks and evaluate incoming submissions. Backend APIs handle authentication, task management, and submission workflows, while the frontend provides an interactive and responsive user interface.

1.4 Results / Findings

The developed system successfully establishes a functional micro-volunteering platform. It enables seamless interaction between volunteers and organizations through task posting, browsing, and submission features.

The role-based system ensures proper access control, allowing organizations to manage tasks while volunteers focus on contributions. The submission workflow, including file uploads and review mechanisms, provides a realistic task validation process. Additionally, the dashboard system allows users to track their activities and progress effectively.

The platform demonstrates improved accessibility to volunteering opportunities and simplifies the process of contributing skills in a structured and efficient manner.

1.5 Technology / Method Used

The platform is implemented as a full-stack web application. The frontend is developed using React.js, providing a dynamic and responsive user interface. The backend is built using Node.js and Express.js, handling API requests and business logic.
MongoDB is used as the database to store user information, tasks, and submission records. Authentication is implemented using JWT (JSON Web Tokens) to ensure secure access control. File uploads are managed using middleware, enabling users to submit task-related content. The overall architecture ensures scalability, modularity, and efficient data handling.

1.6 Expected Outcome & Benefits

The system is expected to provide a streamlined platform for micro-volunteering, making it easier for individuals to contribute their skills without long-term commitments. Volunteers benefit from gaining experience, recognition, and skill development, while organizations receive timely support for specific tasks.

Key benefits include:

- Efficient task distribution and management
- Increased participation in volunteering activities
- Transparent contribution tracking and validation
- Recognition through points and badges
- Scalable and user-friendly system design

Overall, the platform promotes community engagement and enables meaningful contributions through a structured digital ecosystem.

1.7 Original Contribution Statement

This project introduces a unified platform for micro-volunteering by integrating role-based task management, submission workflows, and contribution tracking within a single system. Its originality lies in combining short-term skill-based volunteering with a structured review and recognition mechanism, addressing gaps in existing platforms.

TECHNICAL PROJECT DETAILS

2.1 Programming Language(s)

The project is primarily developed using **JavaScript** for both frontend and backend development. HTML and CSS are used for structuring and styling the user interface.

2.2 Frameworks / Libraries / Platform

The frontend is built using React.js, enabling a dynamic and responsive user interface. The backend is developed using Node.js with Express.js to handle server-side logic and API routing. Additional libraries include Axios for API communication and Multer for handling file uploads.

2.3 Database / Storage Used

The system uses MongoDB, a NoSQL database, to store user data, tasks, and submission records. Data is managed through Mongoose, which provides schema-based modeling for efficient database operations.

2.4 Operating System / Development Environment

The application is developed in a cross-platform environment using Visual Studio Code as the primary code editor. It can run on operating systems such as Windows, macOS, or Linux. Development and testing are performed locally using Node.js runtime.

2.5 External APIs / SDKs Used

The project primarily uses internally developed REST APIs. No major third-party external APIs are integrated at this stage. However, JWT (JSON Web Token) is used for secure authentication and authorization.

2.6 Project Version / Release Number

The project is primarily developed using **JavaScript** for both frontend and backend development. HTML and CSS are used for structuring and styling the user interface.

SOURCE CODE AND DOCUMENTATION

3.1 Source Code Location

The complete source code of the project *SkillBridge* is organized into two main modules: frontend and backend. The frontend is developed using React.js and is located in the client directory, while the backend is developed using Node.js and Express.js and is located in the server directory. Both modules are maintained within a single project folder structure.

Github link : https://github.com/eden221B/SkillBridge

3.2 Number of Source Files

The project consists of multiple source files divided across frontend and backend components. The frontend includes approximately 10 –15 files covering components, pages, services, and styling, while the backend includes approximately 10 –15 files for models, controllers, routes, and middleware. In total, the project comprises around 20 – 30 source files.

3.3 Documentation

The project includes the following documentation:

- Project Report detailing system design, architecture, and implementation
- Source code files (controllers, routes, and services) that implicitly define API structure
- MongoDB schema definitions implemented using Mongoose models (User, Task, Completion)
- README file with setup instructions can be included for project execution

3.4 Screenshots

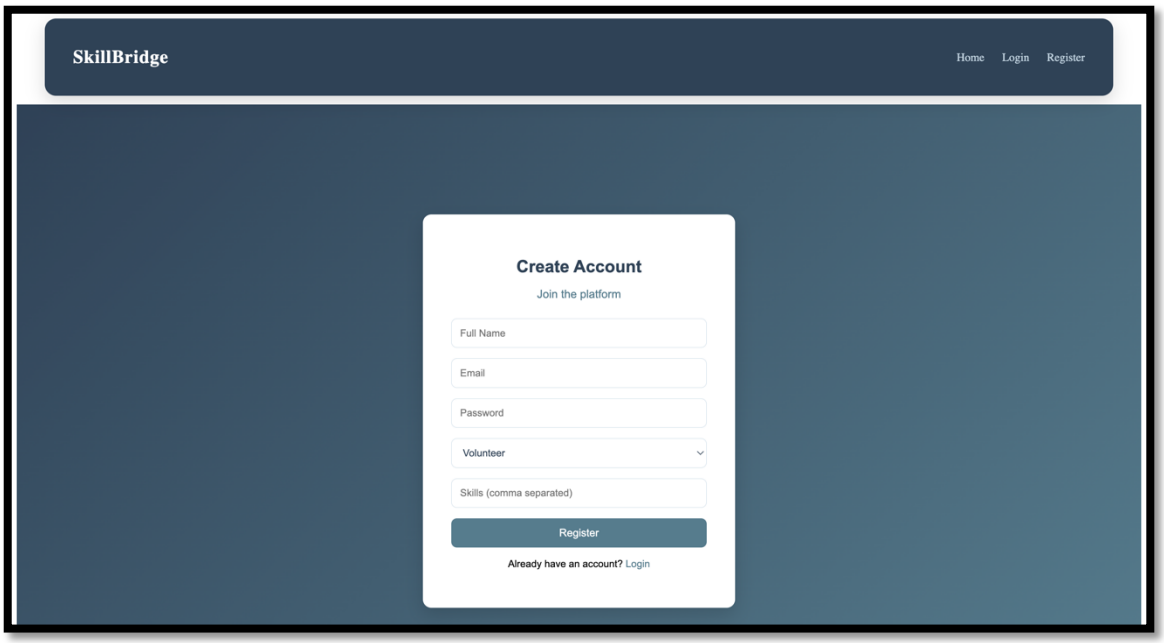

Fig 3.1: Registration Page

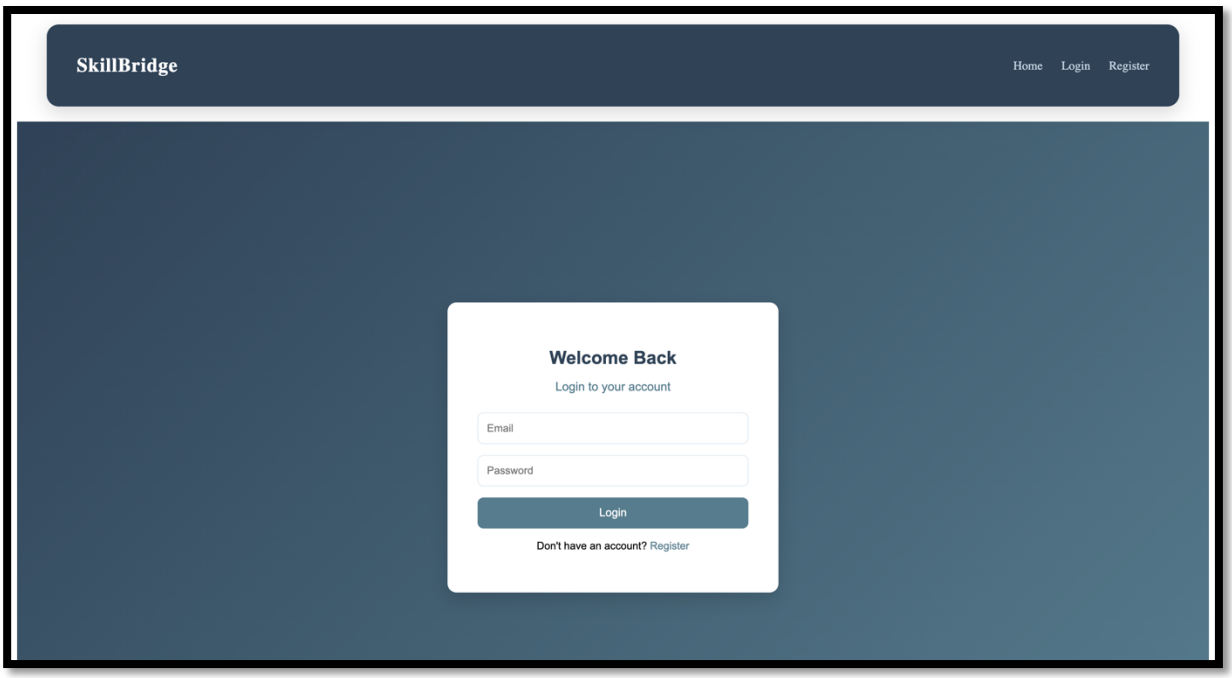

Fig 3.2: Login Page

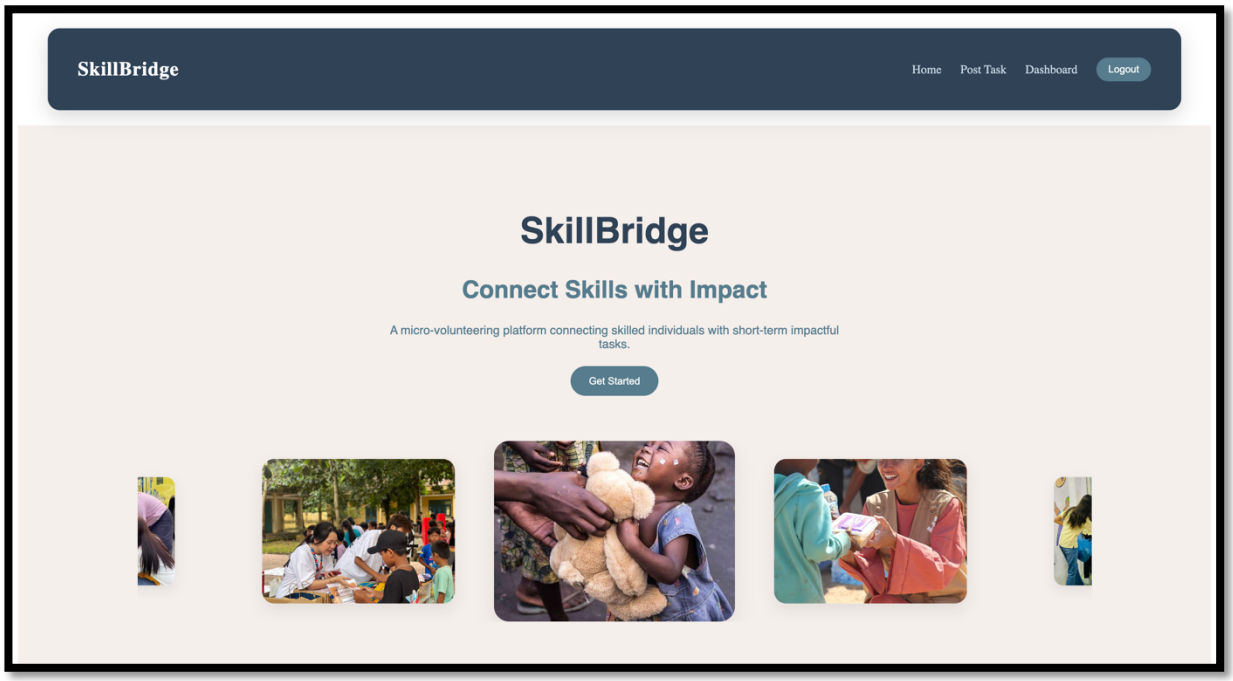

Fig 3.3: Home Page

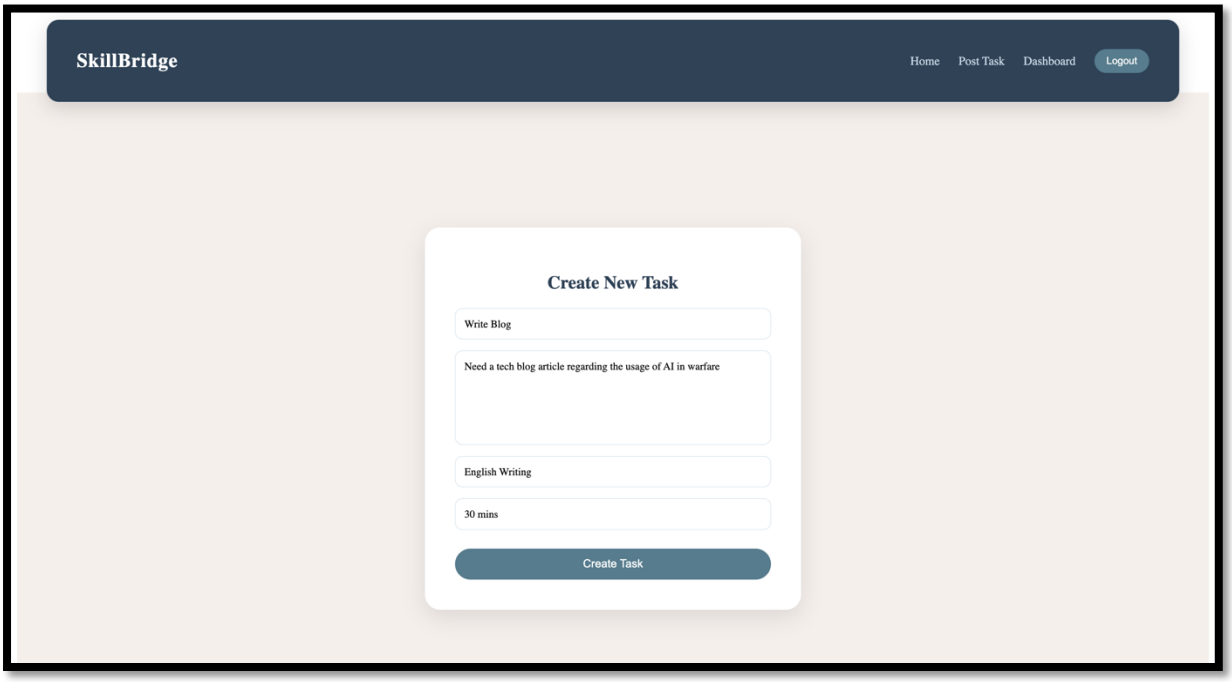

Fig 3.4: Post Task Page

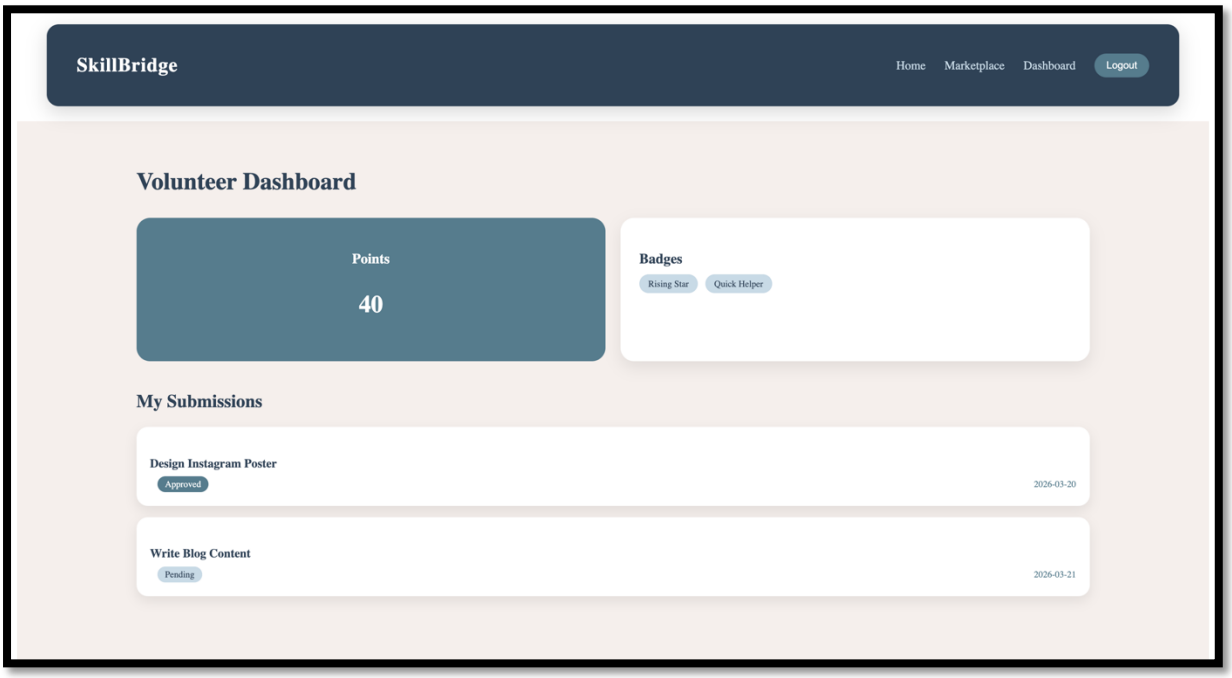

Fig 3.5: Volunteer Dashboard

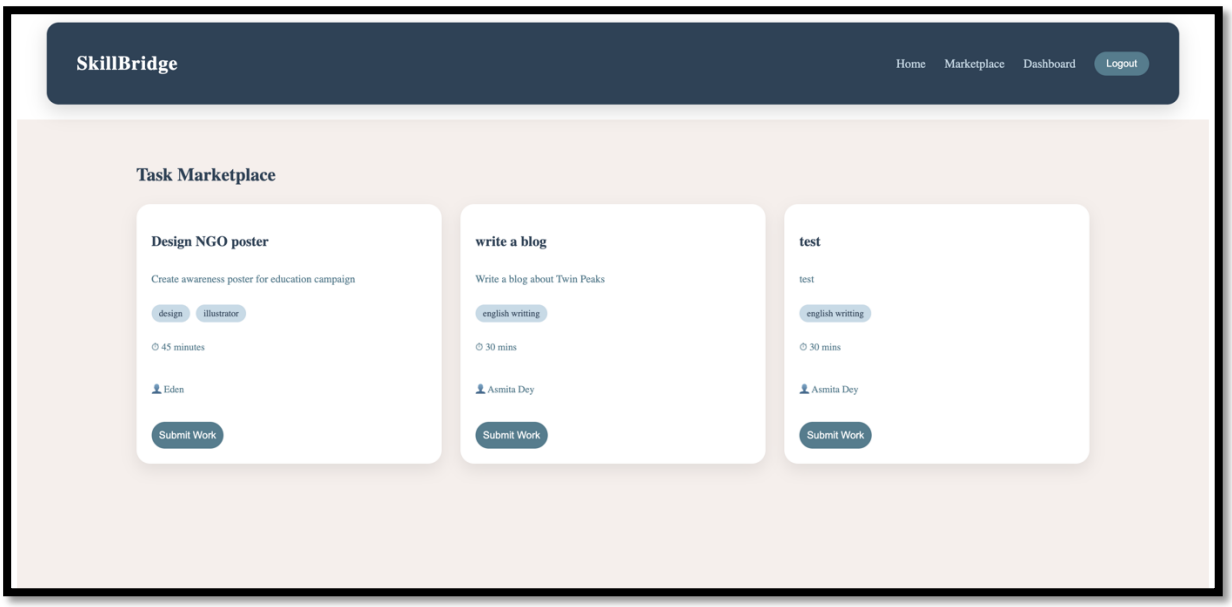

Fig 3.6: Task Marketplace

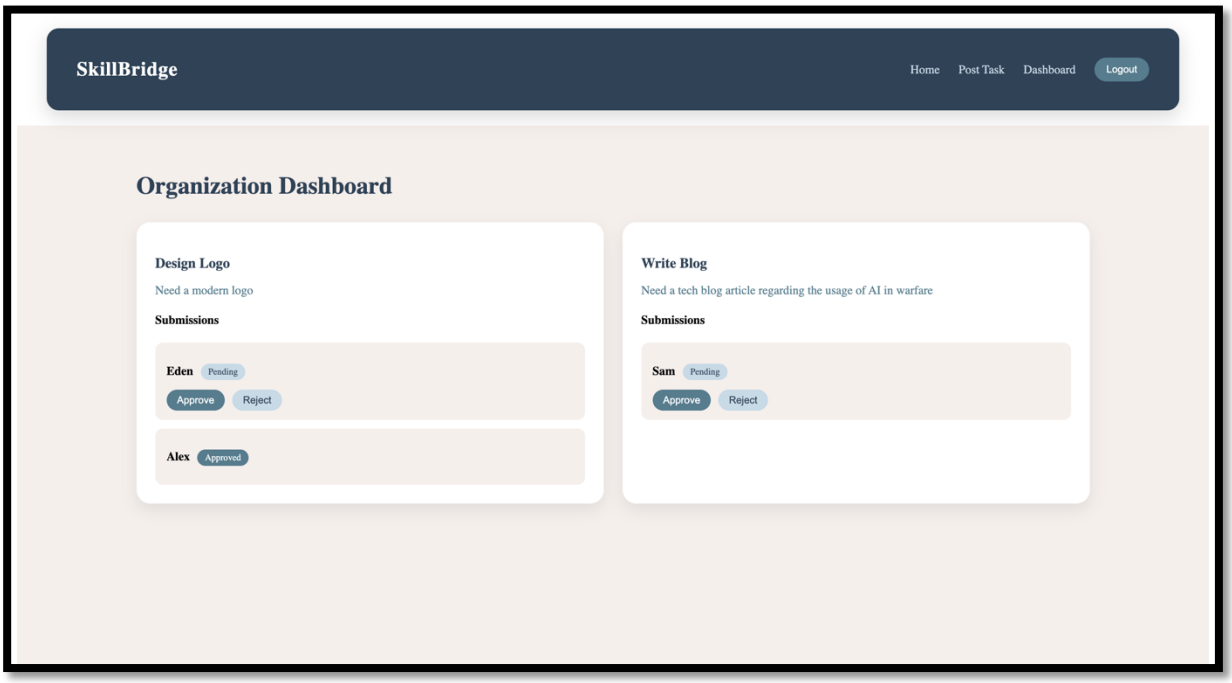

Fig 3.7: Organization Dashboard

3.4 Executable/Build Files

The project is a web-based application and does not require standalone executable files. It can be executed locally using the following setup:

- Frontend: Run using npm run dev (Vite development server)
- Backend: Run using npm run dev (Node.js server with nodemon)
- Database: MongoDB (local or cloud-based)

The system requires a Node.js environment and a configured MongoDB database to function properly.

INTELLECTUAL PROPERTY DETAILS

4.1 Declaration of Originality: Yes
4.2 Any Part Reused from Open-Source Projects: No
4.3 Prior Publications / Presentations (if any): No
4.4 Date of First Creation / Development:
4.5 Submitted previously for copyright: No

DECLARATION

I hereby declare that the information provided above is true and correct. The project is original work carried out by the team under the supervision of our mentor.

Student Name & Signature: Asmita Dey

Date : 20 March 2026